

oldest among the list we are going to discuss. The Bourne shell was developed by Stephen Bourne at Bell Labs. It was first released in 1979 as part of the UNIX operating system and in most Linux OSs, the Bourne shell is pre-installed. The command `'echo $0'` prints the current shell. Figure 1 shows that the default shell in my Ubuntu operating system is Bash. The Bourne shell can be invoked with the command `'sh'` on the terminal. We can see that the second execution of the command `'echo $0'` shows that the current shell is now the Bourne shell.

Figure 1 also shows the output of the commands `'pwd'` and `'date'` in the Bourne shell. From Figure 1 we can also see that the command `'exit'` will take you back to the default shell (Bash). An important question is regarding the License under which the Bourne shell is released. Here there is some confusion. Solaris distributes a version of the Bourne shell under the Common Development and Distribution License (CDDL) and FreeBSD distributes a version of the Bourne shell under the BSD license. Though both are free and open source licenses, the confusion remains as to under which license Bourne shell is released at present. Since Bash is well maintained and supported, it is not known if people actually use the Bourne shell as their primary shell of choice.

```
File Edit View Search Terminal Help
# echo $0
bash
# sh
$ echo $0
sh
$ pwd
/home
$ date
Mon Jun 1 13:47:36 IST 2020
$ exit
# echo $0
bash
```

Figure 1: Using the Bourne shell in Linux

GNU Bash

Now let us discuss the default shell for most of the distributions of Linux called GNU Bash. Even though the shell is popularly known as Bash, it was

originally named as *Bourne-again shell*, a pun on the name of the Bourne shell. This is yet another example for funny names being given to GNU software that replaced original UNIX software. Other such examples include the parser generator Bison (replacing Yacc), the spell checker Aspell (in place of Ispell in UNIX) and the recursive acronym of GNU ('GNU's not UNIX!') itself. Bash was developed by Brian Fox, and is compliant with the POSIX standard. It is free and open source software licensed under the GNU General Public License. We have already seen that Bash is the default shell in the Ubuntu operating system. In fact, Bash is so popular, it is the default shell in Linux distributions like Ubuntu, Fedora, CentOS, openSUSE, etc. But as seen by us already, it is very easy to switch from one shell to another. Figure 2 shows some basic commands like `'pwd'` and `'date'` executed on the Bash terminal.

```
File Edit View Search Terminal Help
# echo $0
bash
# pwd
/home
# date
Sun Jun 7 22:34:20 IST 2020
```

Figure 2: Using Bash in Linux

KornShell

Now let us work with another popular shell called KornShell, which was developed by David Korn at Bell Labs. It was originally proprietary software but now is free and open source software under the Eclipse Public License. Because of its early history as proprietary software, there are many open source alternatives to KornShell like mksh, pdksh, etc. Notice that, unlike the Bourne shell and Bash, KornShell is not pre-installed in Ubuntu. But the installation took just one command `'sudo apt install ksh'` in Ubuntu and was very smooth. I am sure the installation of KornShell on other Linux distributions will be equally simple. KornShell is usually invoked with the command `ksh`. Figure 3 shows

some basic commands like `'pwd'` and `'date'` executed on the KornShell terminal. From Figure 3, we can also see that with the execution of the command `'exit'` the default shell Bash is ready again for operation.

```
File Edit View Search Terminal Help
# ksh
$ echo $0
ksh
$ pwd
/home
$ date
Sun Jun 7 23:01:08 IST 2020
$ exit
# echo $0
bash
```

Figure 3: Using KornShell in Linux

The Z shell

Now let us discuss the Z shell, which was developed by Paul Falstad. It is free software released under the MIT License. Till macOS Mojave (Apple's desktop operating system for Macintosh computers) the default shell was Bash, but in the current operating system called macOS Catalina, the Z shell has been chosen as the default option. I believe this is a notable achievement and justifies the inclusion of Z shell in this list. Notice that, like KornShell, Z shell is also not pre-installed in Ubuntu. But once again, the installation took just one command: `'sudo apt install zsh'`. This time also the installation was very smooth. The Z shell is usually invoked with the command `'zsh'`. Figure 4 shows the execution of the commands `'echo $0'`, `'pwd'`, and `'exit'` on the Z shell terminal.

```
File Edit View Search Terminal Help
# zsh
deepu% echo $0
zsh
deepu% pwd
/home
deepu% exit
# echo $0
bash
```

Figure 4: Using the Z shell in Linux

The C shell

The popular C shell was developed by